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Sushi Preference in Japan: A Discrete Choice Analysis

Understanding Consumer Behavior Through DCM — HW03

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What Drives Sushi Choice?

Research Question

- How do price, oiliness, and availability explain sushi preference?
- Do age and East/West region moderate taste for oiliness?
- Can we predict substitution & market scenarios?

Data: SUSHI Dataset

- 5,000 Japanese consumers (Kamishima, 2003)
- **Set A**: 10 items, full ranking (used throughout)
- Attributes: price, oiliness, freq. sold
- Demographics: age, gender, 12 regions

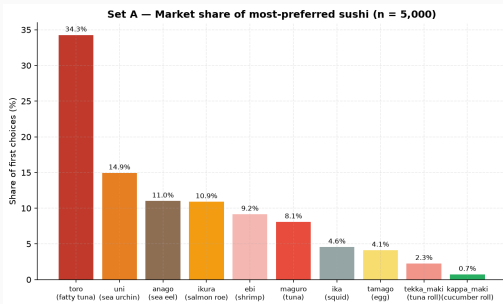


Figure 1: *

Toro dominates (34.3%). Uni, anago, ikura follow.

The Identification Challenge

Critical Design Property

Price, oiliness, and freq_sold are **constant per item** — all 5,000 users see the same values.

⇒ **ASCs and attribute betas cannot coexist** in one specification (perfect collinearity).

Solution: nested chain of models with LR tests

MNL-attributes (3 betas) $\subset$ **MNL-ASC** (9 ASCs)

“How much do
attributes explain?”

“What do the
attributes *miss*?”

↓ **LR test**

M1 (9 ASCs + demographic interactions)

“Who deviates from the average?”

The three headline models are **M1** (interactions), **M2** (Nested Logit) and **M3** (Latent Class); the two baselines above only anchor the LR chain.

M1 — Preferred Model: Age Loves Oil

Specification

$$U_{nj} = ASC_j + \beta_{p \times w} \cdot price_j \cdot west_n + \beta_{o \times a} \cdot oil_j \cdot age_n + \beta_{o \times w} \cdot c$$

Interaction	Coef.	z
$\beta_{price \times west}$	-0.066	-1.46
$\beta_{oil \times age}$	+0.089	5.20
$\beta_{oil \times west}$	-0.058	-0.86

LL = -9,733 $\rho^2 = 0.155$ LR vs MNL-ASC: $p = 10^{-9}$

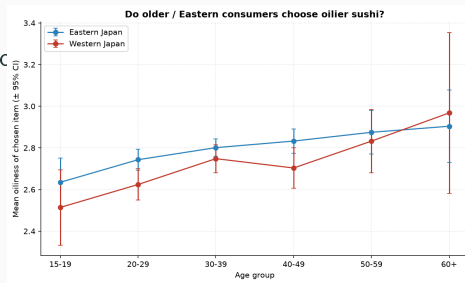


Figure 2: *

Older Eastern consumers choose the oiliest sushi.

Bottom line

Each step up in age group adds +0.089 utils per unit of oiliness — **older consumers consistently prefer richer, fattier sushi.**

East vs. West Japan: A Measurable Divide

Individual interactions: not significant

Both $\beta_{\text{price} \times \text{west}}$ and $\beta_{\text{oil} \times \text{west}}$ have $|z| < 1.96$.

But joint LR test: $p = 0.0004$

The East–West gap is *real* but price and oiliness are too collinear ($\rho = 0.82$) to separate cleanly.

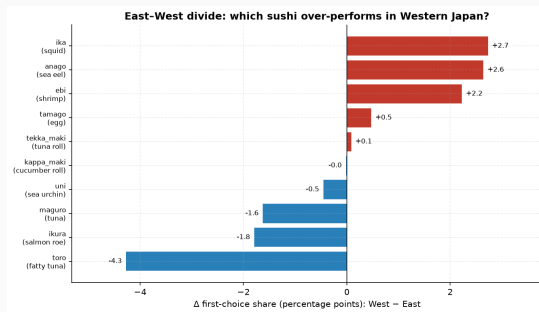


Figure 3: *

Systematic East–West differences in choice shares.

Why it matters

The README of the original dataset *predicts* this East–West divide. Our model *measures* it, but the attributes' collinearity prevents pinpointing the exact channel (price sensitivity vs. oiliness aversion).

M2 — Nested Logit Rejects IIA

Exploded ranking (ranks 1–3)

Each user contributes 3 pseudo-choices:

- Stage 1: all 10 items available
- Stage 2: 9 items (rank 1 removed)
- Stage 3: 8 items (ranks 1–2 removed)

⇒ 15,000 pseudo-observations

Three nests by ingredient

1. akami: maguro, toro, tekka_maki
2. seafood_other: ebi, anago, ika, uni, ikura
3. non_seafood: tamago, kappa_maki

λ parameter	Estimate	t vs. 1
λ_{akami}	0.323	−26.1
$\lambda_{\text{seafood_other}}$	0.342	−6.9
$\lambda_{\text{non_seafood}}$	1.000	(fixed)

LR vs MNL-exploded = 573.5, $p \approx 0$

A free $\lambda_{\text{non_seafood}}$ hits the $\lambda \rightarrow 0$ boundary (degenerate 2-item nest), so it is fixed at 1.

$\lambda < 1$ in both free nests

Substitution is much stronger within nests than MNL predicts.

M3 — Three Latent Consumer Segments (LC-MNL, Apollo)

Latent Class MNL on the exploded ranking; class membership driven by age, region, gender, migration. BIC selects **3 classes** (LL $-28,514$ vs. $-29,446$ with 1 class).

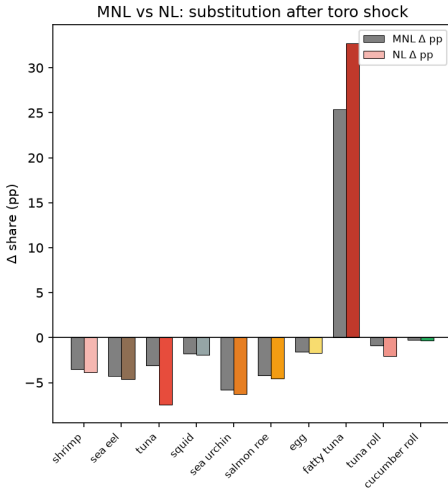
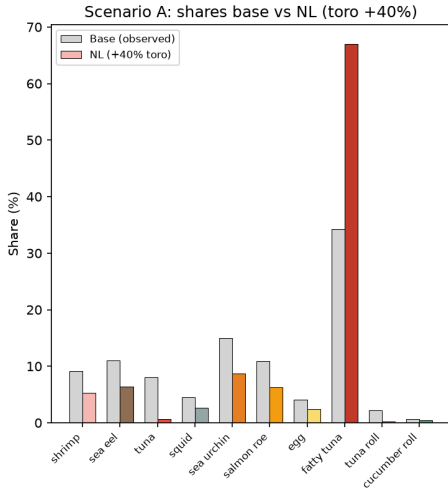
	Premium gourmet	Light taste	Tuna fanatic
Share	33.7%	31.3%	34.9%
Signature items	uni, ikura, toro	ebi (all else < 0)	toro, maguro, tekka
Rejects	egg, rolls	uni (-2.2)	cucumber roll
Profile	oldest, 51% F	youngest, West, 64% F	most male, East

Convergent evidence

The “tuna fanatic” class *is* the akami nest seen as a person type. And uni’s polarization (loved by gourmets at $+1.9$, rejected by the light class at -2.2) explains the rank-stability drift found in robustness checks.

Scenario A: What If Toro Becomes Scarce?

Intervention: Toro price +40% (bluefin tuna quota reduction)

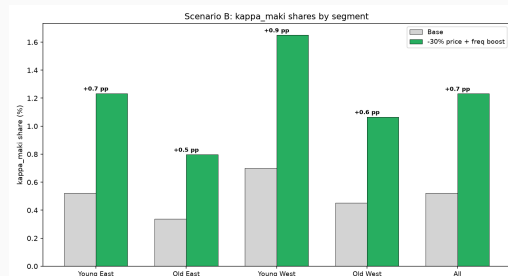


Scenario B: Can Price Make Sushi Healthier?

Intervention: kappa-maki (cucumber roll)

- -30% price
- availability `freq_sold` 0.40 → 0.88 (p75)
- heterogeneity via M1 `age` × `West`

Segment	Base %	CF %	Δ pp
Young East	0.52	1.23	+0.71
Young West	0.70	1.65	+0.95
All	0.52	1.23	+0.71



Health-policy insight

Even -30% price + a big availability push lifts the healthiest item only from 0.5% to 1.2%.

And since $\beta_{\text{price}} > 0$, the *discount hurts* — the gain comes entirely from availability (β_{freq}).

Strategic Recommendations

For Industry

1. **Menu design:** When toro sells out, stock up on maguro and tekka_maki — not proportional restocking (M2, $\lambda_{akami} = 0.32$).
2. **Regional targeting:** Premium fatty items → older Eastern consumers. Light items → younger Western consumers.
3. **Segment offers:** Three latent segments (gourmet / light / tuna-fanatic) are addressable by age, gender and region (M3).

For Public Policy

1. **Don't subsidize healthy sushi:**
 $\beta_{price} > 0$ means cheaper = lower perceived quality.
2. **Boost availability instead:**
 $\beta_{freq_sold} = +2.17$ is the largest coefficient. Make healthier options visible.
3. **Reframe, don't discount:** Market light sushi as premium craftsmanship, not “cheap healthy alternative.”
4. **Cultural preservation:** Regional taste differences are real — leverage identity, not price.

Limitations & Next Steps

Key Caveats

- $\beta_{\text{price}} > 0$: price signals quality in this survey; WTP and price scenarios are **not monetary**.
- $\text{corr}(\text{price}, \text{oiliness}) = 0.82$ in Set A: multicollinearity inflates SEs, so price scenarios are read qualitatively.
- Rank explosion assumes stable preferences across ranking stages.
- Fixed 10-item choice set: strong LR tests, but no generalisation beyond Set A.

Validation & future work

- Apollo (R) cross-validation **completed**: log-likelihoods match the Python engine to ≤ 0.01 across all models.
- Latent Class (3 segments) estimated; a continuous Mixed Logit could capture residual within-class heterogeneity.
- Revealed-preference data for interpretable price effects.

Thank you